BACKGROUND

As dynamic go-getter at a Security Operations Center (SOC), you check the Security Information and Event Management (SIEM) system and find several signature hits for NetSupport Manager RAT from 45.131.214[.]85 over TCP port 443. The activity started on 2026-02-28 at 19:55 UTC.

Using this information, you quickly retrieve a packet capture (pcap) of the traffic from the internal IP address that triggered these alerts. It's all on you now! You're expected to write up an incident report, so someone can track down the infected computer and put a stop to this nonsense!

The characteristics of your environment are:

- LAN segment range: 10.2.28[.]0/24 (10.2.28[.]0 through 10.2.28[.]255)
- Domain: easyas123[.]tech
- AD environment name: EASYAS123
- Active Directory (AD) domain controller: 10.2.28[.]2 - EASYAS123-DC
- LAN segment gateway: 10.2.28[.]1
- LAN segment broadcast address: 10.2.28[.]255

Armed with pcap, you intend to find that infected host.

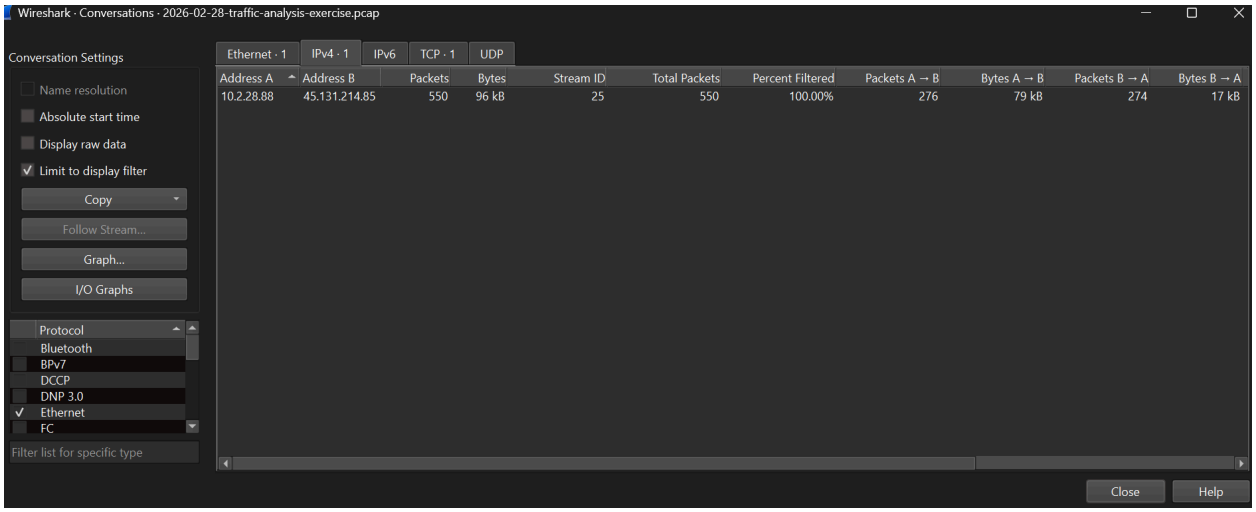
YOUR TASK

For this exercise, answer the following questions for your incident report:

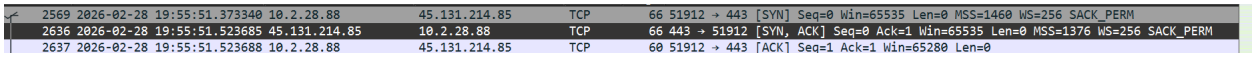
- What is the IP address of the infected Windows client?
- What is the MAC address of the infected Windows client?
- What is the host name of the infected Windows client?
- What is the user account name from the infected Windows client?
- What is the full name of the user from the user account?

Step 1)

Isolate the C2 (command and Control server) -> NetSupport Manager RAT from 45.131.214[.]85 over TCP port 443. Genuine tool, but often used by hackers for remote access. Cross reference packet data with those of infected - check ipv4 tab under statistics → conversations to see possible infected devices.



As shown in figure A. Display filter was used to show a conversation from culprit IP 10.2.28.88, sending 550 bytes via Ipv4 to 45.131.214[.]85. 96kb in total was transferred during the conversation. Suspected compromised host.



As shown in figure A2, a three way TCP handshake can be seen conducted between the C2 server and host device. Client source port is 51912.

None

```
POST http://45.131.214.85/fakeurl.htm HTTP/1.1
User-Agent: NetSupport Manager/1.3
Content-Type: application/x-www-form-urlencoded
Content-Length: 22
Host: 45.131.214.85
Connection: Keep-Alive
CMD=POLL
INFO=1
ACK=1
HTTP/1.1 200 OK
Server: NetSupport Gateway/1.92 (Windows NT)
Content-Type: application/x-www-form-urlencoded
Content-Length: 69
Connection: Keep-Alive
```

Figure A3 is a HTTP follow stream from packet 2639, once connection has been established a post from the host device can be seen to server, confirming both NetSupport usage and data transfer of

244 bytes. Connection is being kept alive, so several connections are not being made at once, data is encoded. Following appended filter of http then viewing statistics → http → requests for further evidence in figure B.

ip.addr == 10.2.28.88 && nbns → following filter to identify host name. Packet 144 shows host name in following snippet A4.

```
None
144    2026-02-28 19:55:12.212763 10.2.28.88    10.2.28.255   NBNS   110
Registration NB DESKTOP-TEYQ2NR<20>
```

Using ip.addr == 10.2.28.88 && arp (address resolution protocol) we can find the mac address associated with the ip. Following returned no results, adjusted filter to arp and looked manually.

```
None
3      2026-02-28 19:55:08.124544 Intel_b2:4d:ad      Broadcast     ARP    60
Who has 10.2.28.1? Tell 10.2.28.88
```

Possible infected client requesting router IP → looking under arp for packet 3:

```
None
Frame 3: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
Ethernet II, Src: Intel_b2:4d:ad (00:19:d1:b2:4d:ad), Dst: Broadcast
(ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Intel_b2:4d:ad (00:19:d1:b2:4d:ad)
    Sender IP address: 10.2.28.88
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.2.28.1
```

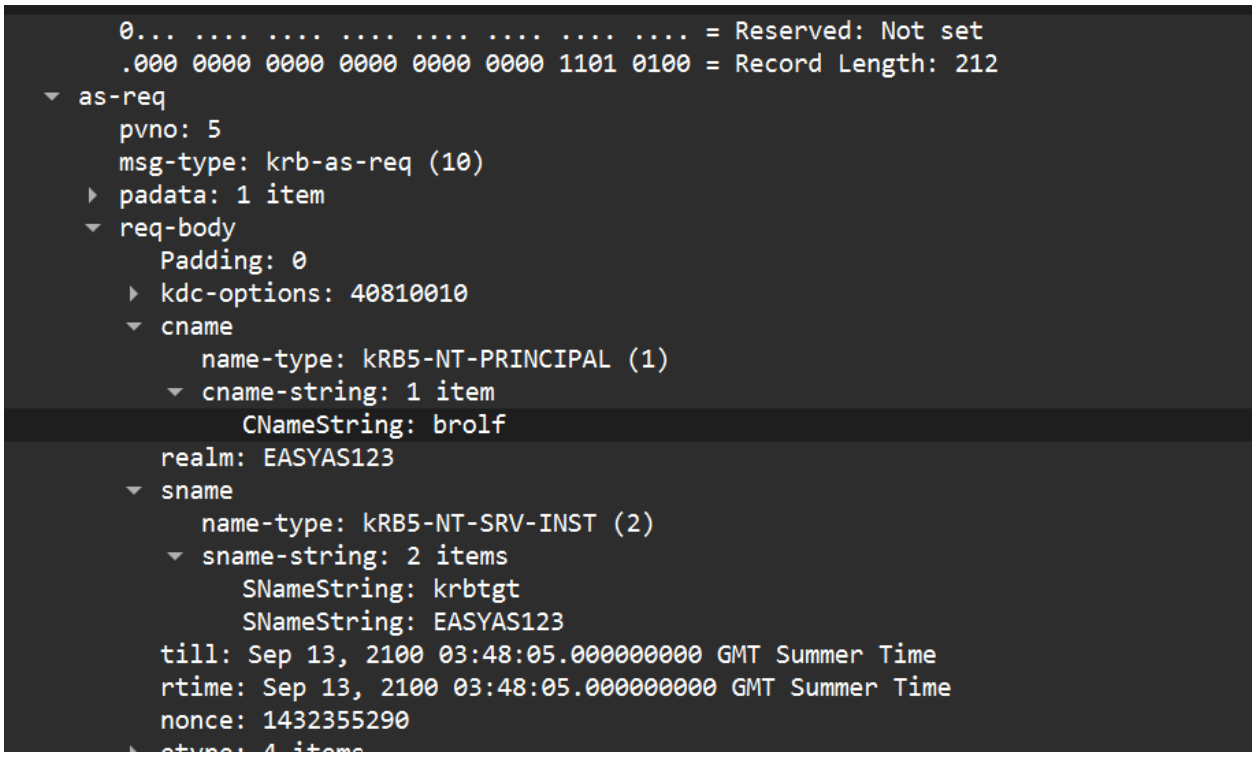
Mac address highlighted in figure A5 above.

Next ip.addr == 10.2.28.88 && smb2 filter will be applied to check any windows file sharing & network operations to identify the user and then subsequently the users full name. This is as these protocol return the windows account name during the authentication protocols.

```
None
70    2026-02-28 19:55:09.694911 10.2.28.88   10.2.28.2    SMB2   859    Session
Setup Request
```

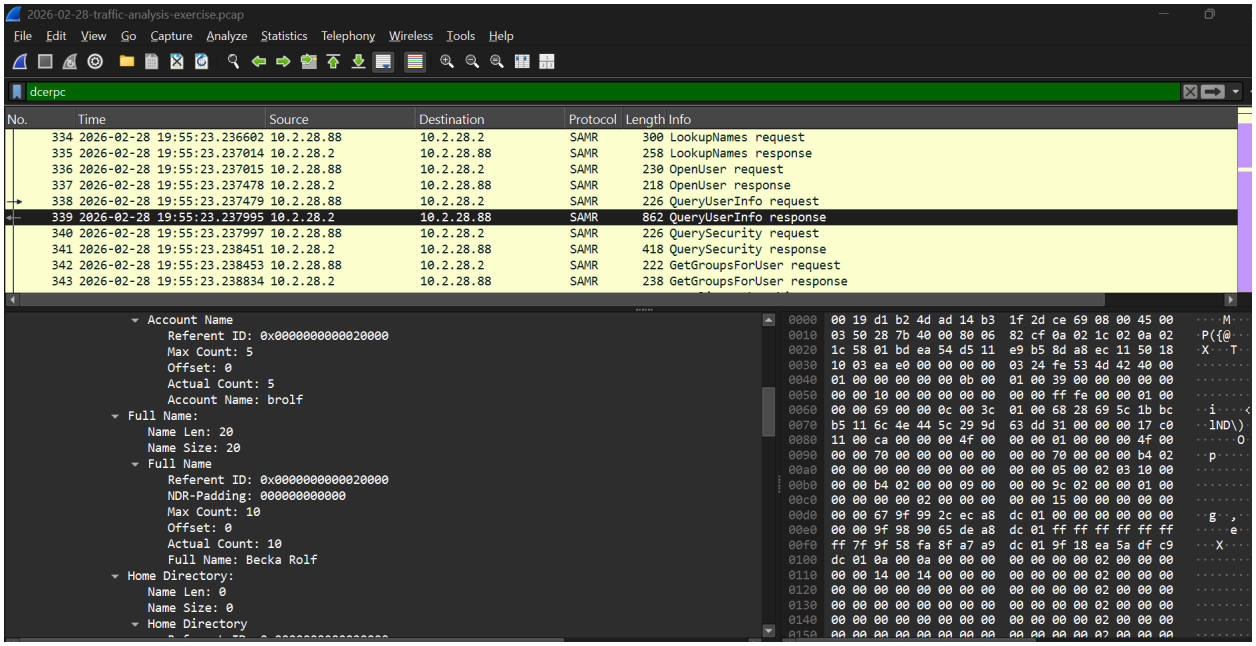
Client establishes an SMB session with a server in figure A6 in packet 70.

ip.addr == 10.2.28.88 && kerberos → to filter fields with the windows authentication protocol, scanning for initial request which shows the clients name in Cname field as shown below in figure A7.



Went through filter _ contains brolf - protocol drsuapi containing encrypted date could not parse DsCrackNames response

SAMR look up protocol with user ‘brolf’ found in frame 335 ‘brolf’ has code RID 1103 as active directory object. Following packet trail down.



Full name as relating to ID and the SAMR QueryUserInfo response is Becka Rolf.